Division properties of exponents and the zero exponent

1 Simplify the following, giving your answers in exponential form.

a $10^8 \div 10^6$ **b** $j^8 \div j^4$ **c** $a^5 \div a^3$ **d** $9q^6 \div (4q^2)$

2 Simplify the following, giving your answers with positive exponents.

a $\dfrac{18^{32}}{18^{23}}$ **b** $\dfrac{a^{10}}{a^6}$ **c** $\dfrac{u^{13}}{u^2}$ **d** $\dfrac{t^7 r^8}{t^5 r^4}$

e $\dfrac{5r^7}{r^4}$ **f** $\dfrac{3j^{13}}{4j^8}$ **g** $\dfrac{-x^{11}}{3x^4}$ **h** $\dfrac{9j^9 k^8}{4j^7}$

i $\dfrac{3g^8}{2g^5 h^6}$ **j** $\dfrac{9j^5 k^7}{j^3 k^4}$ **k** $\dfrac{j^8 k^9}{7j^7 k^8}$ **l** $\dfrac{3m^9 n^4}{8m^8 n^2}$

m $\dfrac{4p^3 q^9}{2p^2 q^7}$ **n** $\dfrac{2p^9 q^6}{6p^3 q^3}$ **o** $\dfrac{3j^6 k^4}{2j^4 k^2}$ **p** $\dfrac{y^6}{(4y)^2}$

q $\dfrac{6r^3 s^6}{2r^2 s^4}$ **r** $\dfrac{-9x^{13}}{3x^4}$ **s** $\dfrac{x^6}{4x^4}$ **t** $\dfrac{54x^{11}}{6x^6}$

3 For the following, write the integer value or the term that should go in the space.

a $a^9 \div a^{\square} = a^7$ **b** $15j^{14} \div (\square) = 5j^7$

c $x^{23}y^{10} \div (\square) = x^{15}y^5$ **d** $\square \div (g^4 h^3) = g^{16}h^5$

e $63x^{18} \div \square = 9x^{14}$ **f** $x^{21}y^8 \div \square = x^{14}y^3$

g $\dfrac{8r^3 s^8}{\square} = 2rs^2$

4 Simplify the following, giving your answers in exponential form:

a $\dfrac{(n^8 r^5)^5}{(n^4 r)^5}$ **b** $\dfrac{30x^{13}y^{26}z^{21}}{3x^{11}y^{15}z^6}$

c $\dfrac{15x^{16}y^{25}z^{13}}{-3x^{14}y^{11}z^2}$ **d** $(m^{12})^9 \div (m^4)^2$

e $(-240u^{32}) \div (-8u^9) \div (-5u^{12})$ **f** $\dfrac{y^7 \cdot y^5}{y^3 \cdot y^2}$

g $\dfrac{10p^6 \cdot 3p^{10}}{15p^2}$ **h** $\dfrac{(2x^4 y^0)^2}{x^6}$

i $\dfrac{6\,(w^2 v)^5 \cdot 27y^{17}v^7}{(3w^5 v^3)^2 \cdot 2y^6 v}$ **j** $m^9 \div m^5 \cdot m^4$

k $p^{18} \div p^8 \div p^5$ **l** $\dfrac{(x^3)^2}{x^3}$

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m $\dfrac{45b^6}{(3b)(5a)}$

n $\dfrac{210p^{22}}{7p^8} \div (5p^7)$

5 Fill in the missing value to complete the pattern.

$k^5 = 1 \cdot k \cdot k \cdot k \cdot k \cdot k$

$k^4 = 1 \cdot k \cdot k \cdot k \cdot k$

$k^3 = 1 \cdot k \cdot k \cdot k$

$k^2 = 1 \cdot k \cdot k$

$k^1 = 1 \cdot k$

$k^0 = \square$

6 Fill in the missing exponent to make the following equations true.

a $p^{12} \div p^{12} = p^{\square}$

b $\dfrac{b^{\square}}{b^{11}} = b^0$

c $\dfrac{w^7}{w^{\square}} = w^0$

7 Evaluate:

a 741^0

b $(-983)^0$

c $(2 \cdot 13)^0$

8 Simplify:

a $18a^0$

b q^0

c $11a^0$

d $(6a)^0$

e $\left(a^0\right)^{79}$

f $9 \cdot \left(15x^6\right)^0$

g $\left(3m^4\right)^3 \cdot mq^0$

h $\dfrac{45x^7y^7}{9x^7y^2}$

Additional questions

9 When simplifying the expression $\dfrac{a^5 \cdot a^4}{a^2}$, Chad's first line of work was $a^3 \cdot a^2$.
Is Chad's work correct? Explain.

10 When simplifying the expression $\dfrac{a^5 + a^4}{a^2}$, Clarice's first line of work was $a^3 + a^2$.
Is Clarice's work correct? Explain.

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